

Oxygen Delivery and Consumption

Imagine that you walk into a typical intensive care unit (ICU). You look around and note what you see: in one room a 78-year-old man with urosepsis and septic shock, in the next room a 65-year-old woman with a chronic obstructive pulmonary disease exacerbation, in the next a 55-year-old man with cirrhosis and a gastrointestinal bleed, in the next a 45-year-old man with non-ST-elevation myocardial infarction and intermittent runs of ventricular tachycardia...

The list goes on and on.

What do all of these patients and, frankly, any patient admitted to the ICU have in common? They all have very different pathology, medical histories, and modalities of support that they are requiring.

However, all of these patients are in the ICU because they have a deficiency or are at risk of having a deficiency in their ability to deliver oxygen. This is tremendously important to recognize about our patients – the common reason for critical care interventions is a deficiency in the delivery of oxygen. The better we understand the concept of how oxygen is delivered, the better position we are in to define and support this delivery of oxygen.

HOW IS OXYGEN DELIVERED IN THE BODY?

Have you ever wondered why it is that the tissue that surrounds the heart and lungs is not preferentially oxygenated, compared to the tissue in the periphery (Fig. 1.1)?

At first glance, this is what you might expect. After all, the tissue surrounding the heart and lungs is located centrally and has the nearest proximity to oxygen when it first comes into the body. Indeed, if oxygen was distributed throughout the body by diffusion, then this would be the case.

What do I mean by diffusion? Diffusion is the movement of molecules down their concentration gradient from high concentration to low concentration.

Think of how oxygen enters the body in the lungs – you take a breath in and air enters through your nose/mouth, down your trachea, across your bronchi, all the way to the alveoli. At this point the oxygen *diffuses* into the blood, going from an area of high concentration (the alveoli) to an area of lower concentration (the blood) (Fig. 1.2).

For many organisms such as bacteria, amoebae, and some jellyfish, this process of diffusion is the primary mechanism for the distribution of oxygen. However, all higher levels of function/life are dependent on a more efficient process for oxygen delivery.

Why is diffusion inefficient at delivering oxygen? Let's start by sketching the potential relationship of oxygen delivery by diffusion. If we were to do so, we might expect something that looks like (Fig. 1.3), with a relatively straight line that slopes up, with more delivery of oxygen at higher concentrations and less delivery of oxygen at lower concentrations.

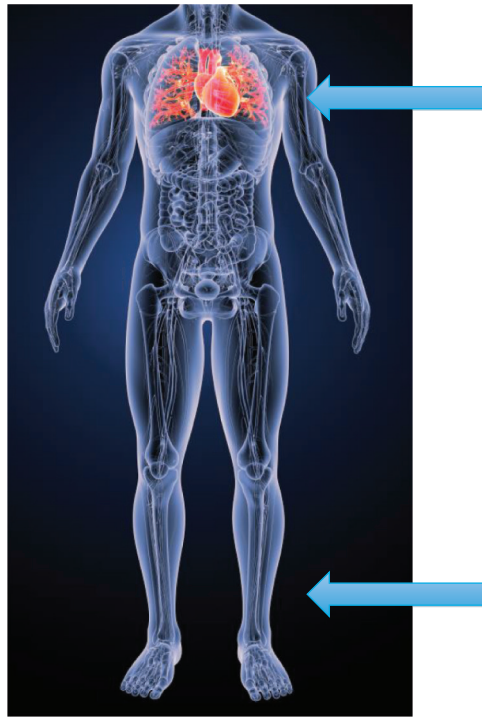


FIG. 1.1 Maintaining oxygenation of both the central and peripheral tissues. (From [Sciencepics/Shutterstock.com.](#))

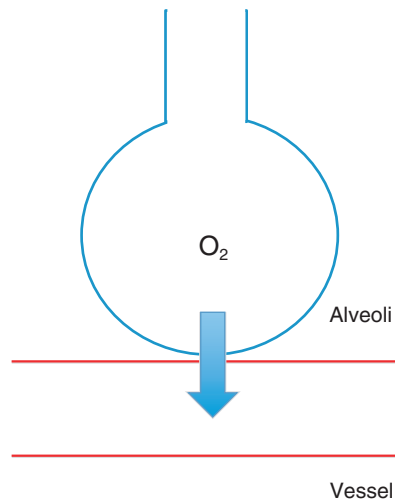


FIG. 1.2 The alveolar-capillary interface as an example of oxygen diffusion in the body

If this were how oxygen were delivered, you would not anticipate much efficiency, as tissues that were present at higher oxygen concentrations (the right side of the line) would have the majority of O_2 delivered and tissues that were present at lower oxygen concentrations (the left side of the line) would have less oxygen delivered. Fortunately, there is a much more efficient means of delivering oxygen.

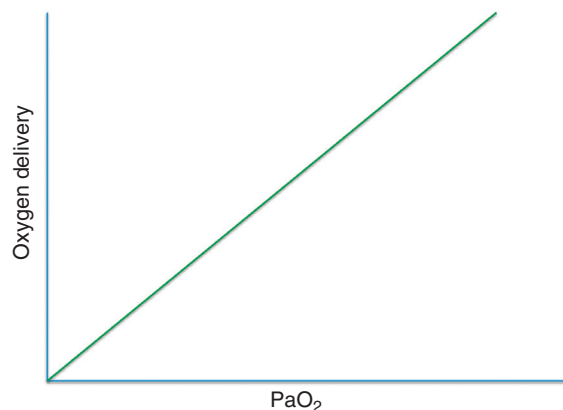


FIG. 1.3 Oxygen delivery by diffusion

WHAT ALLOWS FOR EFFICIENT DELIVERY OF OXYGEN THROUGHOUT THE BODY?

If you were inventing a way to ideally deliver oxygen throughout the body, you would want to have a system that would **bind** preferentially to oxygen, when there is an abundance of oxygen and **dump** preferentially, when there is less presence of oxygen. Returning to our graph, you may ideally expect something like the gray lines below, where there is a lower inflection point, allowing for less delivery/higher binding at high concentrations of oxygen and an upper inflection point, allowing for more delivery/higher dumping at lower concentrations of oxygen (Fig. 1.4).

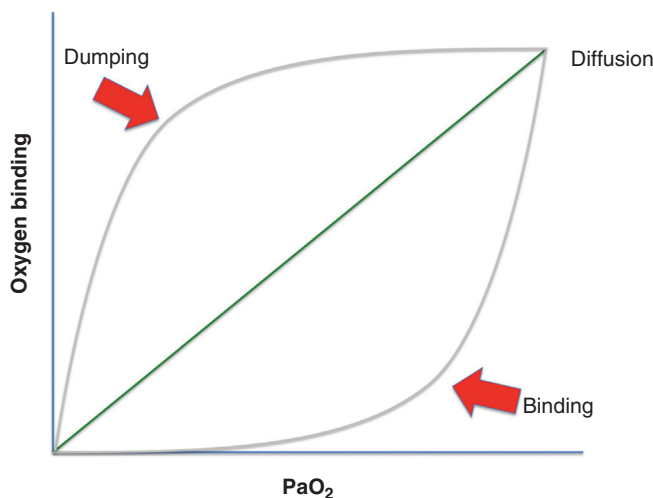


FIG. 1.4 Making oxygen delivery more efficient

Any system that features these two inflection points would be vastly superior and allow for a much more efficient means of delivering oxygen. Fortunately, this is not just an arbitrary thought exercise but, rather, describes the molecule that lies at the center of oxygen delivery – hemoglobin!

HOW DOES HEMOGLOBIN ALLOW FOR MORE EFFICIENT OXYGEN DELIVERY?

Remember the ideal molecule for delivery of oxygen has two properties:

1. Preferentially **bind** when there is an abundance of oxygen
and

2. Preferentially **dump** when there is a scarcity of oxygen

Let's explore how hemoglobin is able to accomplish both of these goals.

Hemoglobin and oxygen binding

Hemoglobin is a dynamic molecule, which carries with it the ability to fold its proteins and change conformation. Its proteins form a bond with oxygen allowing up to four oxygen molecules to bind to each hemoglobin molecule (Fig. 1.5). However, that is only half the story.

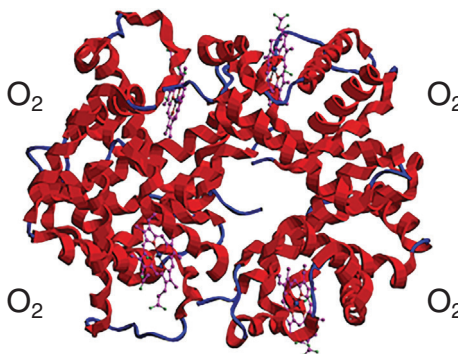


FIG. 1.5 Hemoglobin and oxygen binding. (From [Raimundo79/Shutterstock.com](https://www.shutterstock.com/user/Raimundo79).)

Rather than simply binding to oxygen, the hemoglobin molecule carries with it the ability to change its structure based on how much oxygen is bound to it. If a hemoglobin molecule comes into contact with an oxygen molecule, it will bind as noted earlier. However, once it is bound to the first oxygen molecule, hemoglobin changes conformation in such a way that it will bind more strongly to the second O_2 that comes along than the first. Hemoglobin then changes again – the third O_2 is more likely to bind than the second, and the fourth O_2 is more likely to bind than the third.

You can see that this property causes a positive feedback loop, so that as more O_2 comes in contact with more hemoglobin, the amount bound not only increases, but increases in an exponential fashion.

The effect of this is a higher degree of oxygen binding at higher oxygen concentrations (Fig. 1.6).

Hemoglobin and oxygen dumping

Remember, hemoglobin binding in the presence of oxygen abundance is only half the story. The other half is that hemoglobin must also dump oxygen in the periphery, where there is less oxygen available.

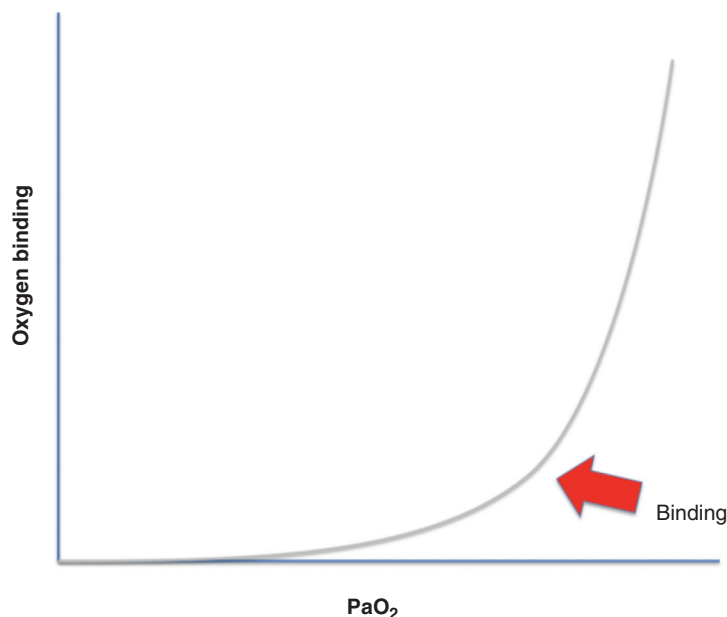


FIG. 1.6 Preferential oxygen binding at high levels of P_{aO_2}

Let's now explore this property of hemoglobin.

The hemoglobin proteins can exist in both a tensed form and a relaxed form. Imagine it as a rubber bowl that can be stretched causing its contents to be spilled out. When hemoglobin exists in the tensed form, oxygen spills out and is dumped for delivery/consumption.

Ideally this tense form would exhibit itself more preferentially in the periphery, which is exactly what happens! This taut form is induced in the presence of higher temperature, lower pH, and higher CO_2 – all conditions common in the periphery (Fig. 1.7A). Thus, each hemoglobin molecule is more likely to be in the tense state and thus dump its oxygen contents in the periphery where there is likely to be less oxygen available (Fig. 1.7B).

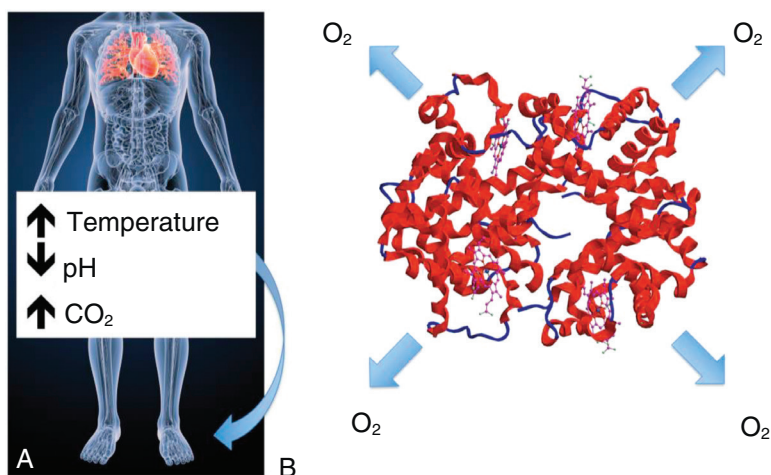


FIG. 1.7 Oxygen conformational changes in periphery. (A, From [Sciencepics/Shutterstock.com](#).; B, From [Raimundo79/Shutterstock.com](#).)

This allow for a system where hemoglobin holds onto oxygen at higher concentrations, only to dump oxygen at lower concentrations (Fig. 1.8).

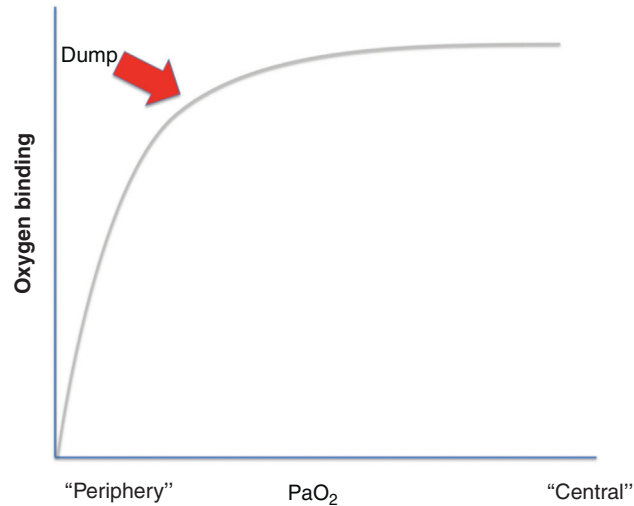


FIG. 1.8 Preferential oxygen dumping at periphery

Taken together, we have the properties of both the lower and the upper inflection points as illustrated below in Fig. 1.9.

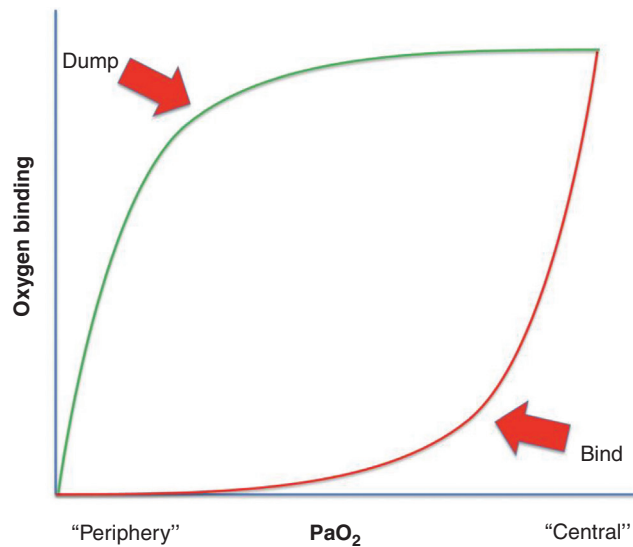


FIG. 1.9 The dual function of hemoglobin - central binding and peripheral dumping

When taken together, these lower and upper inflection points manifest themselves as the sigmoid shape, which is observed in the relationship of hemoglobin binding to oxygen concentration, otherwise known as the hemoglobin dissociation curve (Fig. 1.10).

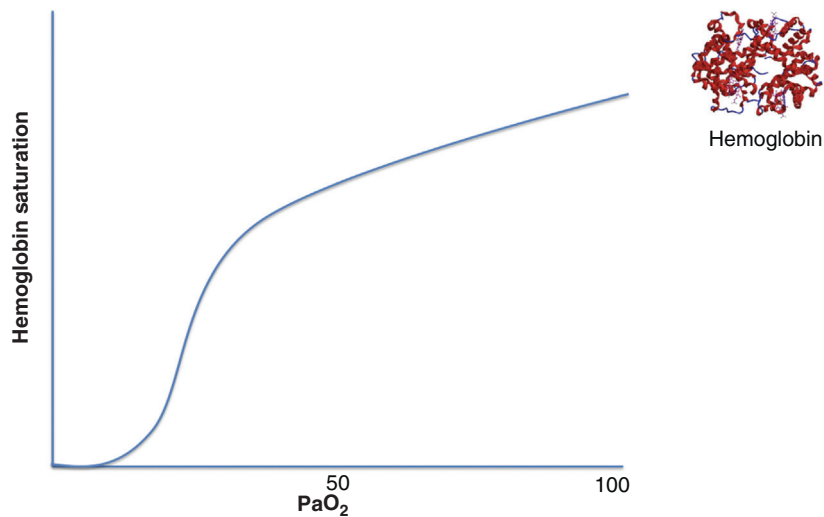


FIG. 1.10 The oxygen hemoglobin dissociation curve. (Modified from [Raimundo79/Shutterstock.com.](#))

Since hemoglobin has the ability to change its conformation, this curve has the ability to shift based on the conditions present. Higher CO_2 , lower pH, and higher temperature cause this curve to shift to the right, meaning that in the heart/lungs, hemoglobin is more likely to bind, while in the periphery, it is more likely to dump (Fig. 1.11).

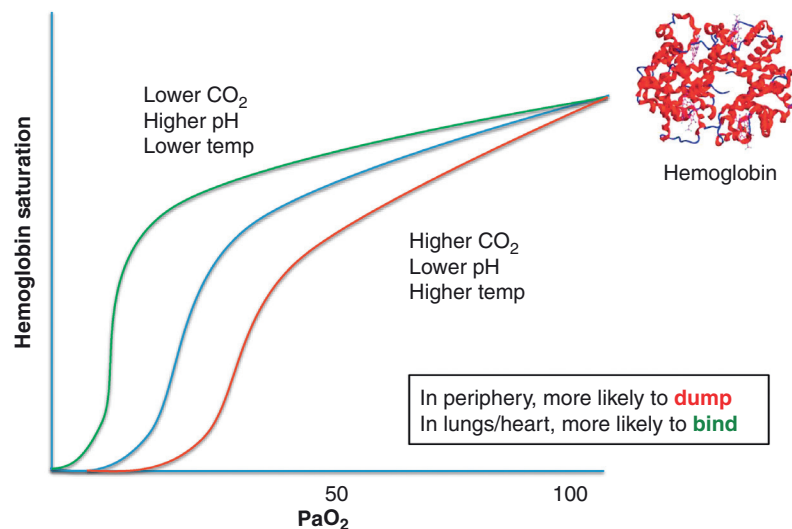


FIG. 1.11 Shifting of the hemoglobin oxygen dissociation curve. (Modified from [Raimundo79/Shutterstock.com.](#))

WHAT IS THE SIGNIFICANCE OF ALL OF THIS WHEN IT COMES TO DELIVERY OF OXYGEN?

This is all so important because the result is that oxygen can be delivered more efficiently. How much more efficiently? Seventy to eighty times more efficient!

Let me say that again, because I think it is the essential point. Because of the efficiency of the hemoglobin molecule, oxygen can be delivered with 70–80 times more efficiently than if by diffusion alone.

Besides instilling a new healthy respect for what hemoglobin is able to do, it also says something about oxygen delivery – the more we can leverage this efficiency, the better we can optimize oxygen delivery, especially when we are in a compromised state.

The new paradigm – oxygen delivery

Understanding oxygen delivery is the ultimate responsibility of anyone taking care of a patient in a critical care setting. The better we can understand, quantify, and optimize the delivery of oxygen, the more equipped we will be for taking care of our patients, even if they are the most compromised patients in the hospital.

How is oxygen delivery defined?

The temptation is that if we want to provide more oxygen delivery, we just provide more oxygen right? Just crank up the O_2 on the wall and we are all set? Wrong! Remember, we have a molecule at our disposal that is 70–80x more efficient than if we were to just rely on diffusion alone. So how do we leverage the efficiency of hemoglobin?

Let's use a common analogy to illustrate this point.

If you have to transport boxes of goods from point A to point B, and you have a truck that is the most efficient carrier of these goods, what are the ways that you can optimize the delivery of these boxes? We should be able to quickly reason that there are only three ways of doing this (Fig. 1.12A):

1. Maximize the load on these trucks.
2. Put more trucks on the road.
3. Have the trucks drive faster.

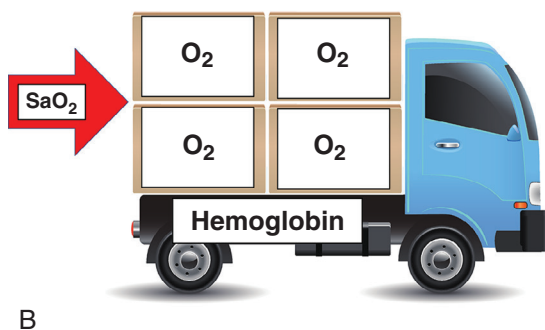


A

FIG. 1.12A The role of hemoglobin in oxygen delivery. (A, From [Bannosuke/Shutterstock.com](https://www.shutterstock.com/user/Bannosuke).)

Let's now extend this analogy to think about oxygen. Now, instead of boxes of goods, let's substitute oxygen, such that the parameters become (Fig. 1.12B):

1. Maximally load up these trucks → increase oxygen saturation.
2. Put more trucks on the road → increase hemoglobin.
3. Have the trucks drive faster → increase cardiac output.



B

FIG. 1.12B The role of hemoglobin in oxygen delivery. (B, From [Bannosuke/Shutterstock.com](#).)

These variables are the basis of the delivery of oxygen equation, which can be written out as:

$$DO_2 = 1.34 \times SaO_2 \times Hb \times CO + PaO_2 \times 0.003$$

How does this apply for your patient?

Let's say you walk into your ICU and see two different patients, Patient A and Patient B. Actually, you see Patient B, but you are *shown* Patient A.

That is because everyone is flagging you to look at Patient A. The respiratory therapist wants you to look at the blood gas that was just drawn, where the PaO_2 is marked as severely abnormal. The bedside nurse is nervously showing you her calculations of cardiac output, which is only calculated to be 6 L per minute.

Patient B meanwhile is off to the side, with few people concerned about his clinical data. The data of Patients A and B are as follows:

Patient A: PaO_2 40 mmHg, Hb 15 g/dL, CO 6 L

Patient B: PaO_2 100 mmHg, Hb 7.5 g/dL, CO 8 L

Knowing what you know now, which patient is more efficiently delivering oxygen?

By doing some simple math, you can see that Patient A is actually delivering more oxygen than Patient B. This is because in this example his hemoglobin concentration is twice as high, allowing him to leverage the efficiency of hemoglobin to a higher degree (Fig. 1.13).

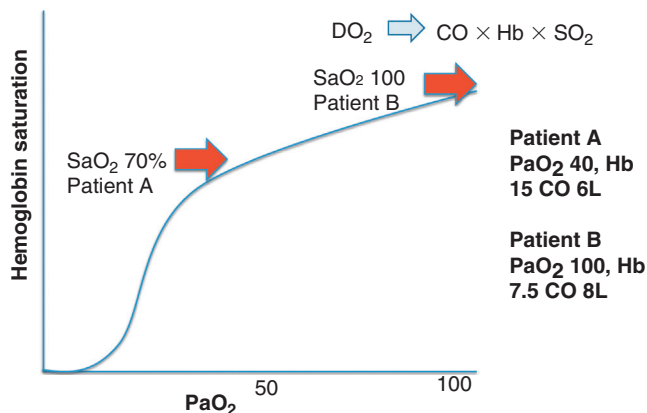


FIG. 1.13 Delivery of oxygen based on different clinical characteristics

HOW IS PAO_2 DIFFERENT FROM SAO_2 ?

Let's differentiate between SAO_2 (oxygen saturation of hemoglobin) and PAO_2 (concentration of oxygen in the blood). Both are often used as a surrogate for how well we are oxygenating, so much so that they are often interchangeable. But you can quickly start to see how different they are!

PAO_2 refers to that partial pressure of oxygen, the dissolved amount of oxygen in the blood. SAO_2 refers to the overall saturation of hemoglobin. While it is true that higher PAO_2 will correspond to a higher hemoglobin O_2 saturation (think of the oxygen-hemoglobin dissociation curve) you can see why they are very different, when it comes to what matters the most to the cells of the body – the delivery of oxygen!

The PAO_2 of the blood in and of itself only allows for delivery of oxygen via diffusion – a much less efficient process. Which is why its contribution to DO_2 is multiplied by a modifier of 0.03 in the DO_2 equation.

Doesn't PAO_2 just correspond to how saturated the hemoglobin molecule is, and therefore the PAO_2 ?

Are you justified in increasing the FiO_2 whenever the blood gas machine flags an abnormal PAO_2 ? Maybe, maybe not.

Remember, the curve shifts depending on what is going on with your patient (Fig. 1.14).

A curve that is shifted to the left (as illustrated by the *green curve*) has a corresponding hemoglobin saturation that is much higher than a curve that is shifted to the right (the *red curve*). And as we will shortly see, in oxygen delivery as in life, there is a cost for every increase in support that we provide.

What ultimately matters to your patient is the DO_2 . No exceptions!

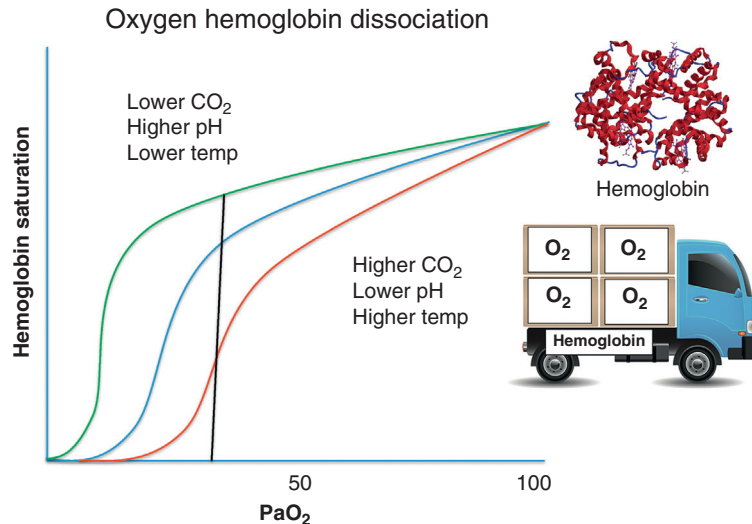


FIG. 1.14 Oxygen hemoglobin dissociation and oxygen delivery. (Modified from [Raimundo79/Shutterstock.com](#), [Bannosuke/Shutterstock.com](#).)

THE ROLE OF OXYGEN CONSUMPTION

Ok, maybe with one exception.

The other side of the delivery of oxygen is the consumption of oxygen. While oxygen delivery is paramount, it must always be put into the context of consumption. Oxygen delivery is only as good as it relates to the body's consumption needs (Fig. 1.15).

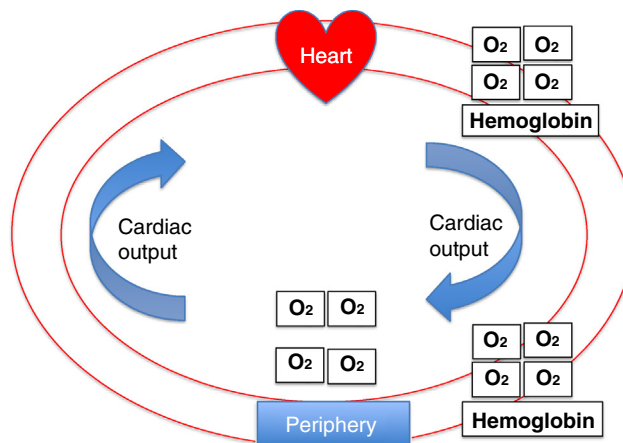


FIG. 1.15 Effect of oxygen consumption in the periphery

To better illustrate this concept, let's imagine ourselves on a treadmill. We are standing on a treadmill at the beginning of our workout. Assuming no deficiencies in our body's ability to deliver oxygen, we are on top of the world, we can stand there indefinitely and feel comfortable that our body is meeting all of its consumption needs.

Now let's start the treadmill and slowly start to increase the speed. What happens? As the treadmill clicks up, from level 1 to 5 to 10, we notice some changes, subtle at first but then becoming more pronounced. Maybe at first we notice that our heart rate is increasing, we start breathing faster, we go from having a full on conversation with the person working out next to us, to speaking in shortened sentences. As the intensity of the treadmill increases higher and higher, we may even start to feel physical pain, distress, our vision blurring, the feeling of oncoming collapse. What is happening here?

What is intuitively happening is that our body is adjusting to match up our DO_2 to our consumption needs, otherwise known as VO_2 .

What does this relationship look like?

If we were to plot the relationship of DO_2 and VO_2 it would look something like this, with every increase in VO_2 (as represented by our increasing levels on the treadmill), represented by progressively higher curves (A to B to C) (Fig. 1.16)

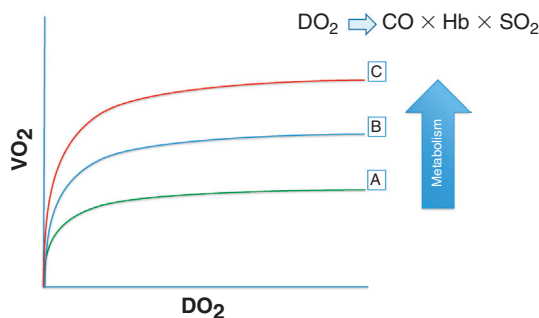


FIG. 1.16 Delivery versus consumption of oxygen at higher levels of consumption

The relationship between DO_2 and VO_2 works in our favor, in that there is a significant amount of redundancy and ability to increase as needed (like we are discovering on our theoretical treadmill). This ability to augment DO_2 (represented by the flat portion of the curve) can readily be increased by 5–6 $\times$ just by increasing our cardiac output. This is what is happening as our treadmill level is increased, and we experience our heart rate increase.

The other redundancy that is built into this relationship is that body only consumes a fraction of the overall oxygen delivered. How can we know and quantify how much we are consuming? Let's explore.

Quantifying VO_2

VO_2 can be difficult to measure directly; therefore we can infer the overall consumption by comparing how much oxygen is being delivered and how much is being returned to the heart. We can envision that as blood is circulated around the body through the periphery, there will be some oxygen that is used and some that is not. The higher the saturation of blood that returns to the heart, the higher the ratio of DO_2 to VO_2 is, as shown in Fig. 1.17.

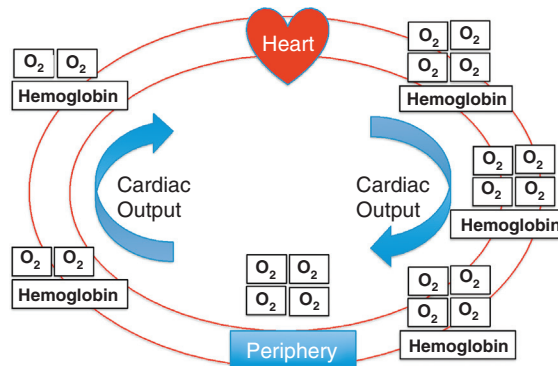


FIG. 1.17 Measuring VO_2

A normal $\text{DO}_2:\text{VO}_2$ ratio might be somewhere between 4:1 and 5:1. This means that we are at any point delivering 4 to 5 times the oxygen as the body needs to consume. This is helpful and allows us to continue to increase our overall VO_2 without the collapse that would be expected at the falloff on the left side of the graph (Fig. 1.18).

However, as we increase the VO_2 from point A to point B, you can observe that the delivery of oxygen has to increase in order to continue to maintain the same $\text{DO}_2:\text{VO}_2$ ratio. If this is not done, the ratio of $\text{DO}_2:\text{VO}_2$ falls from 5:1 to 4:1 or from 4:1 to 3:1 (Fig. 1.18).

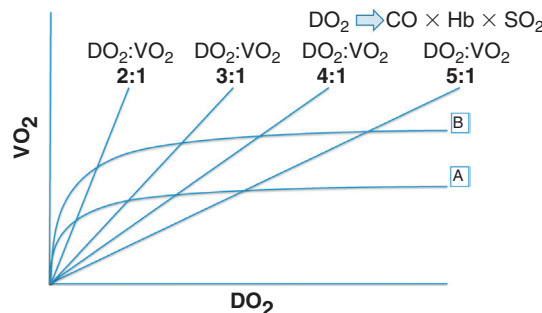


FIG. 1.18 Delivery versus consumption of oxygen

How do we observe this? We can look at the saturation of blood returning to the heart in the form of the venous oxygen saturation. As the $\text{DO}_2:\text{VO}_2$ ratio falls from 5:1 to 4:1, the body then has to consume a higher percentage of the total oxygen delivered, and you would expect the venous oxygen saturation to fall from 80% to 75%. This continues until the $\text{DO}_2:\text{VO}_2$ ratio falls to 2:1, where you might expect the venous oxygen saturation to perhaps fall to 50%. Which, as you can observe on our graph, is coming closer and closer to the falloff point of the graph and overall collapse.

It is this collapse that we are attempting to prevent in every patient admitted to the ICU.

We will now spend the next several chapters understanding how this collapse occurs and what we can do to stop it.

SUGGESTED READING

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